vas.livanos@gmail.com
Website
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**Address:**
3345 Wilshire Blvd
Unit 407
Los Angeles, 90010
United States of America

## Research Interests

- Algorithms under Uncertainty
- Combinatorial Optimization
- Matroid Theory
- Algorithmic Game Theory
- Machine Learning
- Online Mechanism Design

## Languages

- Greek (Native)
- English (Fluent)
- Spanish (Intermediate)
- Italian (Basic)
- French (Basic)

# Vasilis Livanos
## Curriculum Vitae

## Employment

**Mar 2026 - current, University of Southern California**
Post-Doctoral Researcher. Host: Shaddin Dughmi

**Oct 2024 - Apr 2025, EPFL**
Visiting Post-Doctoral Researcher. Host: Andrés Cristi.

**Sep 2024 - current, Archimedes AI**
Affiliated Researcher.

**Jan 2024 - Jan 2026, University of Chile & CMM**
Post-Doctoral Researcher. Hosts: José Correa and José Soto.

## Education

**Aug 2017 - Dec 2023, University of Illinois at Urbana-Champaign**
Ph.D. in Computer Science. Advisors: Ruta Mehta and Chandra Chekuri.
*Thesis:* Optimal Stopping Problems And Combinatorial Optimization Under Uncertainty

**Oct 2011 - Mar 2017, National Technical University of Athens**
Diploma in Electrical and Computer Engineering. Advisor: Dimitris Fotakis.
*GPA: 9.2/10 (Top 2%) - Excellent.*

## Awards and Honours

**2023, *Dissertation Travel Award*, University of Illinois at Urbana-Champaign**
Awarded for the completion of the PhD dissertation.

**2022, *NeurIPS Scholar Award*, NeurIPS '22**
Awarded to provide financial assistance to graduating PhD students.

**2020, *Outstanding Teaching Assistant Award, Lifetime*,**
**University of Illinois at Urbana-Champaign**
Awarded based on ranking by students and recommendation by faculty.

**2018 - 2022, *List of Teachers Ranked as Excellent*,**
**University of Illinois at Urbana-Champaign**
Ranked "Outstanding" based on students' ratings on five occasions.

**2017 - 2018, *Andrew and Shana Laursen Fellowship*,**
**University of Illinois at Urbana-Champaign**
Awarded to provide meaningful assistance in the recruitment of top graduate students
to the Department of Computer Science.

**2012 - 2016, *John S. Latsis Foundation Undergraduate Scholarship*,**
**National Technical University of Athens**
Awarded continuously for achieving excellence during undergraduate studies.

**2011, *Great Moment for Education Scholarship*, Eurobank EFG**
Awarded for being first among all high school students in the national final examinations.

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## Talks and Presentations

**2026, Talk -** ***Threshold Dynamics and Correlated Prophet Inequalities***
27th ACM Conference on Economics and Computation (EC), Rome, Italy.
Criteo AI, Paris, France.
Workshop on Optimization and Algorithms (WOA), Viña del Mar, Chile.

**2025, Talk -** ***Minimization I.I.D. Prophet Inequality via Extreme Value Theory: A Unified Approach***
26th ACM Conference on Economics and Computation (EC), Stanford, CA, USA.
Drexel University, Philadelphia, PA, USA.
UC Irvine, Irvine, CA, USA.

**2025, Talk -** ***Matroid Secretary via Labeling Schemes***
26th Conference on Integer Programming and Combinatorial Optimization (IPCO), Baltimore, MD, USA.
Summer Workshop on Combinatorial Optimization, Santiago, Chile.
SUPSI, Lugano, Switzerland.

**2024, Talk -** ***Improved Mechanisms and Prophet Inequalities for Graphical Dependencies***
INFORMS Annual Meeting, Seattle, USA.

**2024, Tutorial -** ***An Overview of Optimal Stopping Theory***
UC Irvine, Irvine, USA.

**2024, Talk -** ***Oracle-Augmented Prophet Inequalities***
51st EATCS International Colloquium on Automata, Languages and Programming (ICALP),
Tallinn, Estonia.

**2024, Talk -** ***Minimization is Harder in the Prophet World***
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2024, Alexandria, USA.
EPFL, Lausanne, Switzerland.
Google, Zurich, Switzerland.
ETH, Zurich, Switzerland.
University of Bonn, Bonn, Germany.

**2023, Talk -** ***Combinatorial Optimization under Uncertainty***
2nd Drafting Workshop in Discrete Mathematics and Probability, Budapest, Hungary.

**2022, Poster -** ***Simple and Tight Greedy Online Contention Resolution Schemes***
36th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, USA.

**2022, Talk -** ***Combinatorial Optimization under Uncertainty and Prophet Inequalities***
Eötvös Loránd University, Budapest, Hungary.
Karlsruhe Institute of Technology, Karlsruhe, Germany.
Max Planck Institute for Informatics, Saarbrücken, Germany.

**2022, Talk -** ***How to Choose a Partner Using Math***
2022 PhD Intern Conference (PIRC), Google, USA.

**2021, Talk -** ***On Submodular Prophet Inequalities and Correlation Gap***
14th Symposium on Algorithmic Game Theory (SAGT), Aarhus, Denmark.

**2021, Poster:** ***On Submodular Prophet Inequalities and Correlation Gap***
22nd Conference on Economics and Computation (EC), Budapest, Hungary.

## Scientific Visits

**Jan. 2025, 1-month visit at University of Chile**
Host: Jose Correa

**Sep. 2023 - Nov. 2023, 2-month visit at Georgia Institute of Technology**
Host: Sahil Singla

**Feb. 2023, 1-month visit at Eötvös Loránd University**
Host: Kristóf Bérczi

**Jan. 2023, 1-month visit at University of Chile**
Host: Jose Correa

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# Professional Activities and Service

***Program Committee Member***
WINE '26, EC '26, WINE '25, FSTTCS '25, APPROX '25, EC '25, EC '24

***Referree***

- *Journals:* SIAM Journal of Computing, Journal of Computer and System Sciences,
  Journal of Applied Probability, Theoretical Computer Science, Mathematics of
  Operations Research, Algorithmica, Mathematical Programming, Transactions in
  Algorithms, Operations Research Letters.
- *Conferences:* STOC '27, '26, '24, '23, '18 - STACS '27, '24, - ITCS '27, '25, '24 -
  SODA '27, '26, '25, '24, '23 - IPCO '26, '24, '23, '22 - LATIN '26 - FOCS '24 - EC '22, '20, '18 - ESA '23 - SOSA '23 - ICALP '22, '21, '19 - WINE '21, '18 - WWW '18 - SAGT '18.

**Jan. 2025, University of Chile, *committee member***
M.Sc. Thesis of Jorge Hewstone

**EC'22 Mentoring Workshop, *Mentor***

# Teaching

**Jan 2018 - Dec 2022, *Teaching Assistant*, University of Illinois at Urbana-Champaign**

- "CS 583 Approximation Algorithms" (Fall 2021).
- "CS 580 Algorithmic Game Theory" (Fall 2019, Fall 2022).
- "CS 473 Algorithms" (Spring 2018).
- "CS 374 Introduction to Algorithms" (Fall 2018, Spring 2019, Spring 2020, Spring 2021, Spring 2022).

**Mar 2012 - Feb 2017, *Teaching Assistant*, National Technical University of Athens**

- "Algorithms and Complexity" (Fall 2016).
- "Cryptography and Number Theory" (Fall 2016).
- "Introduction to Computer Science" (Spring 2013 - Spring 2016).
- "Programming Techniques" (Spring 2012 - Spring 2016).
- "Introduction to Programming" (Fall 2012 - Fall 2016).

# Industry Experience

**May 2023 - Aug 2023, *Research Intern*, Google, Mountain View**

**May 2022 - Aug 2022, *Software Engineering Intern*, Google, Mountain View**

**Jun 2021 - Dec 2022, *Software Engineer (Contractor)*, Public Benefit Enterprise of Ancient Olympia**

**Jun 2015 - Sep 2015, *Software Engineer (Contractor)*, Proini News**

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## Working & Submitted Papers

**[5]** ***Batched Pandora's Box***
with Shaddin Dughmi, Yusuf Kalayci and Aditya Prasad

**[4]** ***Separating Measures by Hyperplanes***
with José Correa, Svenja Griesbach, El-Farouk Harb, Felix Hommelsheim, Reda Jlibene and Eilon Solan

**[3]** ***Optimal Contention Resolution from Samples***
with Shaddin Dughmi and Pranav Nuti

**[2]** ***The Strong Secretary Conjecture is True for Linear Matroids***
with Kristóf Bérczi, Shaddin Dughmi, José Soto and Victor Verdugo

**[1]** ***Free-Order Secretary for Two-Sided Independence Systems***
with Kristóf Bérczi, José Soto and Victor Verdugo

## Conference Publications

**[11]** ***The Power of Sampling for Finding Far-Apart Bases in Matroids***
with with Mayank Goswami, Pasin Manurangsi, Arturo Merino, GiBeom Park and Adarsh Srinivasan
*SODA 2027*

**[10]** ***Threshold Dynamics and Correlated Prophet Inequalities***
with José Correa, Maximilian Fichtl, Reda Jlibene, Rida Laraki, Kevin Schewior and Victor Verdugo
*EC 2026*

**[9],** ***On the Informativeness of Moments in Optimal Stopping***
with José Correa, Andrés Cristi, Victor Verdugo and Jiechen Zhang
*STOC 2026*

**[8]** ***Minimization I.I.D. Prophet Inequality via Extreme Value Theory: A Unified Approach***
with Ruta Mehta
*EC 2025*

**[7]** ***Matroid Secretary via Labeling Schemes***
with Kristóf Bérczi, José Soto and Victor Verdugo
*IPCO 2025*

**[6]** ***Improved Mechanisms and Prophet Inequalities for Graphical Dependencies***
with Kalen Patton and Sahil Singla
*EC 2024*

**[5]** ***Oracle-Augmented Prophet Inequalities***
with Sariel Har-Peled and El-Farouk Harb
*ICALP 2024*

**[4]** ***Minimization is Harder in the Prophet World***
with Ruta Mehta
*SODA 2024*

**[3]** ***Simple and Optimal Greedy Online Contention Resolution Schemes***
*NeurIPS 2022*

**[2]** ***(Almost) Envy-Free, Proportional and Efficient Allocations of an Indivisible Mixed Manna***
with Ruta Mehta and Aniket Murhekar
*AAMAS 2022*

**[1]** ***On Submodular Prophet Inequalities and Correlation Gap***
with Chandra Chekuri
*SAGT 2021*

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- French (Basic)

# Journal Publications

**[4]** ***About a Ball Removal Process on Bins***
with José Correa, Marcos Kiwi, Eilon Solan and Ron Solan
*Stochastic Systems*

**[3]** ***Matroid Secretary via Labeling Schemes***
with Kristóf Bérczi, José Soto and Victor Verdugo
*to appear in Mathematical Programming*

**[2]** ***Improved Mechanisms and Prophet Inequalities for Graphical Dependencies***
with Kalen Patton and Sahil Singla
*to appear in Mathematics of Operations Research*

**[1]** ***On Submodular Prophet Inequalities and Correlation Gap***
with Chandra Chekuri
*Theoretical Computer Science*

# Theses

**2023, *Optimal Stopping Problems And Combinatorial Optimization Under Uncertainty*, PhD Thesis**

**2017, *Convergence and Equilibria in Coevolutionary Opinion Formation Games*, Diploma Thesis**